

CBSE Class 12 Physics Question Paper 2023

Set 1 (55/1/1) Solutions

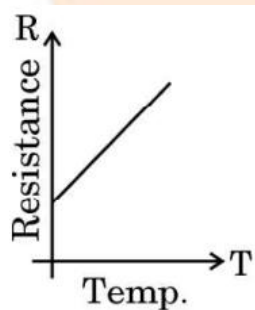
SECTION A

1. A point charge situated at a distance 'r' from a short electric dipole on its axis experiences a force $\vec{F}$. If the distance of the charge is '2r', the force on the charge will be:

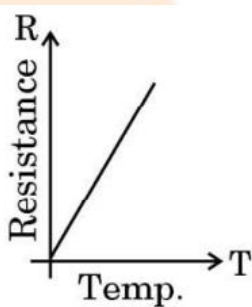
- (a) $\frac{\vec{F}}{16}$ (b) $\frac{\vec{F}}{8}$ (c) $\frac{\vec{F}}{4}$ (d) $\frac{\vec{F}}{2}$

Ans: (b) $\frac{\vec{F}}{8}$

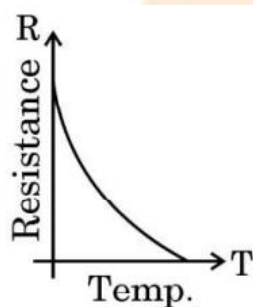
2. For a metallic conductor, the correct representation of variation of resistance R with temperature T is:



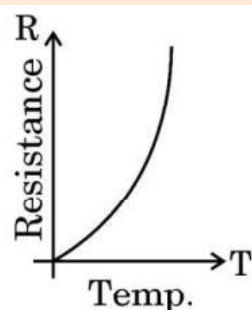
(a)



(b)



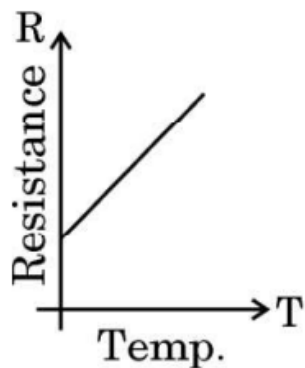
(c)



(d)

Ans: The resistance of a metallic conductor generally increases linearly with temperature. The correct graph is:

(a)



(a)

3. The potential difference across a cell in an open circuit is 8 V. It falls to 4 V when a current of 4 A is drawn from it. The internal resistance of the cell is:

(a) 4Ω

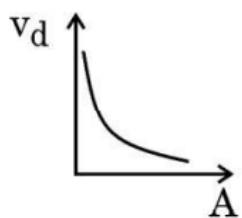
(b) 3Ω

(c) 2Ω

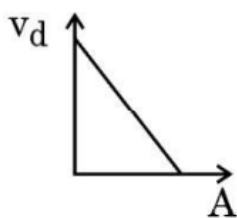
(d) 1Ω

Ans: (d) 1Ω

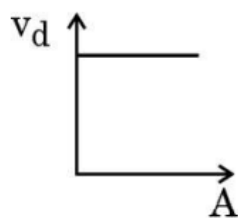
4. A steady current flows through a metallic wire whose area of cross-section (A) increases continuously from one end of the wire to the other. The magnitude of drift velocity (v_d) of the free electrons as a function of 'A' can be shown by:



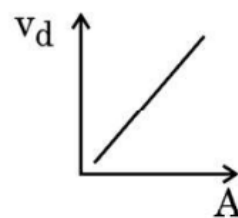
(a)



(b)



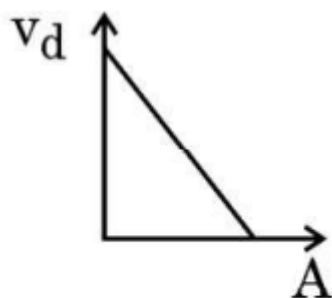
(c)



(d)

Ans: The correct graph is:

(b)



(b)

5. A diamagnetic substance is brought near the north or south pole of a bar magnet. It will be :

- (a) repelled by both the poles.
- (b) attracted by both the poles.
- (c) repelled by the north pole and attracted by the south pole.
- (d) attracted by the north pole and repelled by the south pole.

Ans: (a) repelled by both the poles.

6. A circular coil of radius 8.0 cm and 40 turns is rotated about its vertical diameter with an angular speed of $\frac{25}{\pi} \text{ rad s}^{-1}$ in a uniform horizontal magnetic field of magnitude $3.0 \times 10^{-2} \text{ T}$. The maximum emf induced in the coil is :

- (a) 0.12 V

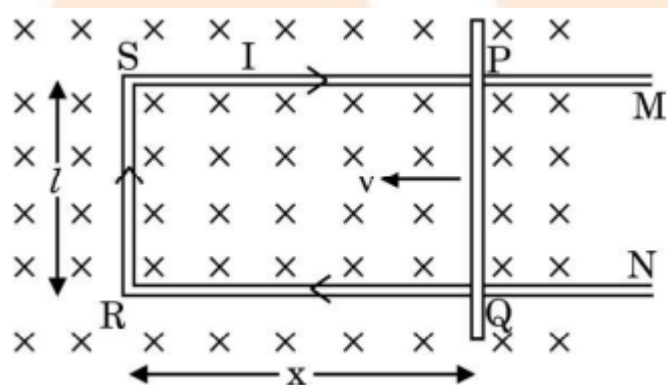
(b) 0.15 V

(c) 0.19 V

(d) 0.22 V

Ans: (c) 0.19 V

7. Figure shows a rectangular conductor PSRQ in which movable arm PQ has a resistance 'r' and resistance of PSRQ is negligible. The magnitude of emf induced when PQ is moved with a velocity $\vec{v}$ does not depend on:



(a) magnetic field ($\vec{B}$)

(b) velocity ($\vec{v}$)

(c) resistance (r)

(d) length of PQ

Ans: (c) resistance (r)

8. In the process of charging of a capacitor, the current produced between the plates of the capacitor is:

(a) $\mu_0 \frac{d\phi_E}{dt}$

(b) $\frac{1}{\mu_0} \frac{d\phi_E}{dt}$

(c) $\epsilon_0 \frac{d\phi_E}{dt}$

(d) $\frac{1}{\epsilon_0} \frac{d\phi_E}{dt}$

where symbols have their usual meanings.

Ans: (c) $\epsilon_0 \frac{d\phi_E}{dt}$

9. For a concave mirror of focal length 'f', the minimum distance between the object and its real image is:

(a) zero

(b) f

(c) 2f

(d) 4f

Ans: (c) 2f

10. The radius of the n^{th} orbit in Bohr model of hydrogen atom is proportional to:

(a) $\frac{1}{n^2}$

(b) $\frac{1}{n}$

(c) n^2

(d) n

Ans: (d) n

11. Hydrogen atom initially in the ground state, absorbs a photon which excites it to $n = 5$ level. The wavelength of the photon is:

(a) 975 nm

(b) 740 nm

(c) 523 nm

(d) 95 nm

Ans: (d) 95 nm

12. The mass density of a nucleus of mass number A is:

(a) proportional to $A^{1/3}$

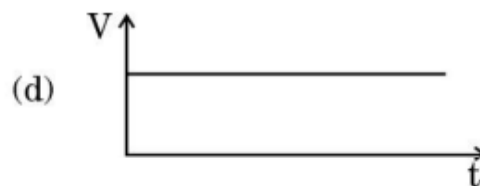
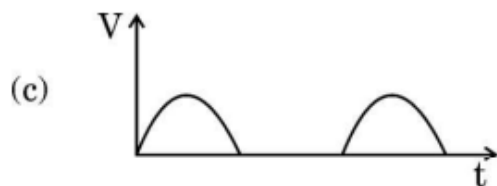
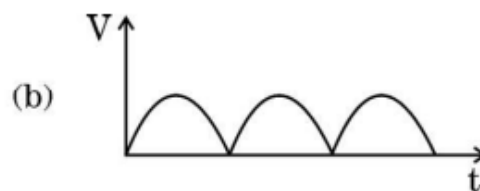
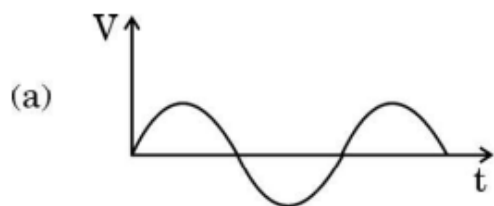
(b) proportional to $A^{2/3}$

(c) proportional to A^3

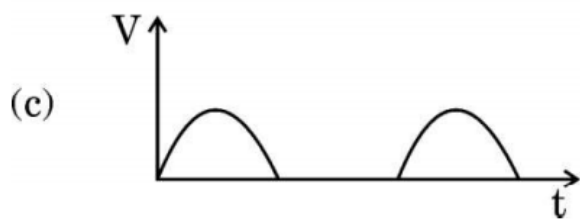
(d) independent of A

Ans: (d) independent of A

13. An AC source of voltage is connected in series with a p-n junction diode and a load resistor. The correct option for output voltage across load resistance will be:



Ans: (c)



14. When an intrinsic semiconductor is doped with a small amount of trivalent impurity, then:

- (a) its resistance increases.**
- (b) it becomes a p-type semiconductor.**
- (c) there will be more free electrons than holes in the semiconductor.**
- (d) dopant atoms become donor atoms.**

Ans: (b) it becomes a p-type semiconductor.

15. In the energy-band diagram of n-type Si, the gap between the bottom of the conduction band E_C and the donor energy level E_D is of the order of:

- (a) 10 eV**
- (b) 1 eV**
- (c) 0.1 eV**
- (d) 0.01 eV**

Ans: (d) 0.01 eV

Questions number 16 to 18 are Assertion (A) and Reason (R) type questions. Two statements are given one labelled Assertion (A) and the other labelled Reason (R).

Select the correct answer from the codes (a), (b), (c) and (d) as given below.

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(c) Assertion (A) is true, but Reason (R) is false.

(d) Assertion (A) is false and Reason (R) is also false.

16. Assertion (A): When a bar of copper is placed in an external magnetic field, the field lines get concentrated inside the bar.

Reason (R): Copper is a paramagnetic substance.

Ans: (d) Assertion (A) is false and Reason (R) is also false.

17. Assertion (A): The phase difference between any two points on a wavefront is zero.

Reason (R): All points on a wavefront are at the same distance from the source and thus oscillate in the same phase.

Ans: (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

18. Assertion (A): Photoelectric effect demonstrates the particle nature of light.

Reason (R): Photoelectric current is proportional to intensity of incident radiation for frequencies more than the threshold frequency

Ans: (b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

SECTION B

19. An alpha particle is projected with velocity $\vec{v} = (3.0 \times 10^5 \text{ m/s})\hat{i}$ into a region in which magnetic field $\vec{B} = [(0.4\text{T})\hat{i} + (0.3\text{T})\hat{j}]$ exists. Calculate the acceleration of the particle in the region. $\hat{i}$, $\hat{j}$, and $\hat{k}$ are unit vectors along x, y and z axis respectively and charge to mass ratio for alpha particle is $4.8 \times 10^7 \text{ C/kg}$. **2**

Ans: To find the acceleration of the alpha particle, we can use the Lorentz force law. The force on a charged particle moving in a magnetic field is given by:

$$\vec{F} = q(\vec{v} \times \vec{B})$$

where:

q is the charge of the particle,

$\vec{v}$ is the velocity of the particle,

$\vec{B}$ is the magnetic field.

The acceleration $\vec{a}$ can be found using Newton's second law:

$$\vec{a} = \frac{\vec{F}}{m}$$

Given that the charge-to-mass ratio $\frac{q}{m}$ for the alpha particle is $4.8 \times 10^7 \text{ C/kg}$, we can directly find the acceleration.

First, calculate the cross product $\vec{v} \times \vec{B}$:

$$\vec{v} = (3.0 \times 10^5 \text{ m/s})\hat{i}$$

$$\vec{B} = (0.4\text{T})\hat{i} + (0.3\text{T})\hat{j}$$

The cross product $\vec{v} \times \vec{B}$ is:

$$\vec{v} \times \vec{B} = (3.0 \times 10^5 \hat{i}) \times (0.4 \hat{i} + 0.3 \hat{j})$$

Using the properties of the cross product:

$$\hat{i} \times \hat{i} = 0 \text{ and } \hat{i} \times \hat{j} = \hat{k}$$

$$\text{We get: } \vec{v} \times \vec{B} = (3.0 \times 10^5) \hat{i} \times (0.4 \hat{i}) + (3.0 \times 10^5) \hat{i} \times (0.3 \hat{j})$$

$$\vec{v} \times \vec{B} = 0 + (3.0 \times 10^5 \times 0.3) \hat{k}$$

$$\vec{v} \times \vec{B} = 9.0 \times 10^4 \hat{k}$$

$$\text{Now, the force } \vec{F} \text{ is: } \vec{F} = q(\vec{v} \times \vec{B}) = q(9.0 \times 10^4 \hat{k})$$

Using the charge-to-mass ratio $\frac{q}{m} = 4.8 \times 10^7 \text{ C / kg}$:

$$\vec{a} = \frac{\vec{F}}{m} = \left(\frac{q}{m} \right) (9.0 \times 10^4 \hat{k})$$

$$\vec{a} = (4.8 \times 10^7) \times (9.0 \times 10^4) \hat{k}$$

$$\vec{a} = 4.32 \times 10^{12} \hat{k} \text{ m / s}^2$$

So, the acceleration of the alpha particle is:

$$\vec{a} = 4.32 \times 10^{12} \hat{k} \text{ m / s}^2$$

20. Consider an induced magnetic field due to changing electric field and an induced electric field due to changing magnetic field. Which one is more easily observed? Justify your answer. **2**

Ans: The induced electric field due to a changing magnetic field is more easily observed compared to the induced magnetic field due to a changing electric field.

Justification:

1. Faraday's Law of Electromagnetic Induction:

Faraday's law states that a changing magnetic field induces an electric field. This is expressed as:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

This phenomenon is commonly observed in everyday applications such as transformers and electric generators. For example, when a magnetic field through a coil changes, it induces an electromotive force (EMF) in the coil, which generates a current if the circuit is closed. This principle is widely used and easily observed in practice.

2. Maxwell-Ampère's Law:

Maxwell-Ampère's law states that a changing electric field induces a magnetic field. This is expressed as:

$$\nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} + \mu_0 \vec{J}$$

While this is a fundamental part of Maxwell's equations, the induced magnetic field due to a changing electric field is generally much weaker and harder to observe directly. The electric displacement current $\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$ is typically small compared to the $\vec{J}$ actual current in practical situations. This makes the induced magnetic field less noticeable.

3. Practical Observations:

- **Induced Electric Fields:** The phenomenon of induced electric fields is directly observable in various practical devices. For example, moving a magnet near a coil or changing the magnetic flux through a loop can easily demonstrate this effect.
- **Induced Magnetic Fields:** The induced magnetic fields due to changing electric fields, while fundamental in theory, are not as easily demonstrated in a typical laboratory setup. They require high-frequency alternating electric fields to produce measurable magnetic effects, which are less common in everyday observations.

21. (a) Using Huygens' principle, draw a ray diagram showing the propagation of a plane wave refracting at a plane surface separating two media. Also verify Snell's law of refraction. 2

Ans: Huygens' Principle

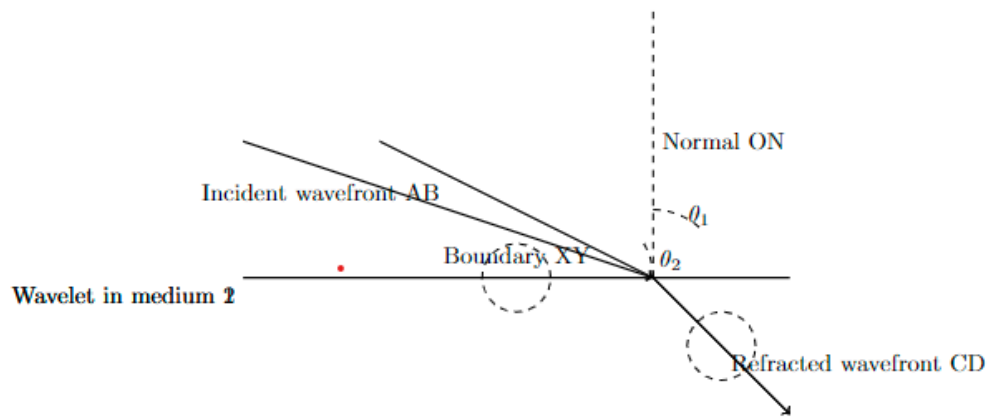
Huygens' principle states that every point on a wavefront is a source of secondary wavelets that spread out in all directions with a speed equal to the speed of the wave. The new wavefront is the envelope of all the secondary wavelets.

Ray Diagram

Let's consider a plane wavefront AB incident on a plane surface separating two media with different refractive indices. The wavefront propagates from medium 1 (refractive index n_1) to medium 2 (refractive index n_2).

1. **Incident Wavefront:** AB is the incident wavefront approaching the boundary at an angle of incidence i .
2. **Refracted Wavefront:** After refraction, the wavefront CD is formed in medium 2 at an angle of refraction r .

Here is the ray diagram illustrating the refraction using Huygens' principle:



Verifying Snell's Law of Refraction

Let θ_1 be the angle of incidence and θ_2 be the angle of refraction.

1. From the geometry of the diagram:

$$\sin\theta_1 = \frac{AB}{AO} = \frac{v_1 t}{AO}$$

$$\sin\theta_2 = \frac{CD}{AO} = \frac{v_2 t}{AO}$$

2. Using the definitions of sine for the respective angles:

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{v_1}{v_2}$$

3. The speed of light in a medium is related to the refractive index n by:

$$v = \frac{c}{n}$$

where c is the speed of light in a vacuum.

4. Substituting the speeds in terms of refractive indices:

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{\frac{c}{n_1}}{\frac{c}{n_2}} = \frac{n_2}{n_1}$$

In this diagram:

- **AB is the incident wavefront.**
- **CD is the refracted wavefront.**
- **θ_1 is the angle of incidence.**
- **θ_2 is the angle of refraction.**
- **O is the point of incidence.**
- **The normals to the wavefronts are shown to illustrate the angles.**

OR

(b) Why is a reflecting telescope preferred over a refracting telescope? Justify your answer giving two reasons.

2

Ans: Reflecting telescopes are preferred over refracting telescopes for two main reasons:

- 1. Elimination of Chromatic Aberration:** Reflecting telescopes use mirrors instead of lenses, which means they do not suffer from chromatic aberration. Mirrors reflect all wavelengths of light equally, resulting in clearer images without color distortion.
- 2. Support for Larger Apertures:** Reflecting telescopes can be built with much larger apertures than refracting telescopes. Mirrors can be supported from the back, allowing for larger and lighter constructions. Larger apertures increase the light-gathering power and resolution, making faint and distant objects more visible.

22. Two coherent monochromatic light beams of intensities I and $4I$ superpose each other. Find the ratio of maximum and minimum intensities in the resulting beam. 2

Ans: When two coherent monochromatic light beams interfere, the resulting intensity depends on the phase difference between them. The maximum and minimum intensities occur due to constructive and destructive interference, respectively.

Given:

- Intensity of the first beam, $I_1 = I$
- Intensity of the second beam, $I_2 = 4I$

Maximum Intensity (Constructive Interference)

Constructive interference occurs when the waves are in phase, and their amplitudes add up. The amplitude A is related to the intensity I by $\mu_0 \epsilon$. Therefore, the amplitudes of the beams are:

$$A_1 \propto \sqrt{I_1} = \sqrt{I}$$

$$A_2 \propto \sqrt{I_2} = \sqrt{4I} = 2\sqrt{I}$$

The resultant amplitude for constructive interference is:

$$A_{\max} = A_1 + A_2 = \sqrt{I} + 2\sqrt{I} = 3\sqrt{I}$$

The maximum intensity is:

$$I_{\min} \propto (A_{\min})^2 = (\sqrt{I})^2 = I$$

Ratio of Maximum to Minimum Intensities

The ratio of the maximum to minimum intensities is:

$$\frac{I_{max}}{I_{min}} = \frac{9I}{I} = 9$$

Therefore, the ratio of maximum to minimum intensities in the resulting beam is 9.

23. The ground state energy of hydrogen atom is -13.6 eV. What is the potential energy and kinetic energy of an electron in the third excited state? 2

Ans: To find the potential energy and kinetic energy of an electron in the third excited state of a hydrogen atom, we need to consider the energy levels of the hydrogen atom.

Ground State Energy

The ground state energy of the hydrogen atom is given as $E_1 = -13.6\text{eV}$

Energy Levels

The energy of an electron in the n-th energy level of a hydrogen atom is given by:

$$E_n = \frac{E_1}{n^2}$$

where n is the principal quantum number.

Third Excited State

The third excited state corresponds to $n = 4$ (since the ground state is $n = 1$, the first excited state is $n = 2$, the second excited state is $n = 3$, and the third excited state is $n=4$).

So, the energy in the third excited state is:

$$E_4 = \frac{E_1}{4^2} = \frac{-13.6\text{eV}}{16} = -0.85\text{eV}$$

Kinetic Energy

The total energy of the electron in the n-th state is the sum of its kinetic energy (K.E) and potential energy (P.E.). For a hydrogen atom:

$$E_n = K.E + P.E$$

It is known that:

$$E_n = -\frac{1}{2} E_n$$

So, the kinetic energy in the third excited state is:

$$K.E = -\frac{1}{2} \times -0.85eV = 0.425eV$$

Potential Energy

The potential energy is given by:

$$P.E = 2 \times E_n$$

So, the potential energy in the third excited state is:

$$P.E = 2 \times -0.85eV = -1.7eV$$

Therefore, in the third excited state (n=4):

- The potential energy of the electron is -1.7 eV.
- The kinetic energy of the electron is 0.425 eV.

24. (a) Differentiate between intrinsic and extrinsic semiconductors.

2

Ans: Intrinsic Semiconductors:

- **Definition:** Pure semiconductors with no added impurities. Examples include pure silicon (Si) and germanium (Ge).
- **Carrier Concentration:** Equal number of electrons and holes since carriers are generated by thermal excitation.

Extrinsic Semiconductors:

- **Definition:** Semiconductors doped with specific impurities to increase conductivity. Examples include silicon doped with phosphorus (n-type) or boron (p-type).
- **Carrier Concentration:** Unequal number of electrons and holes. In n-type, electrons are the majority carriers; in p-type, holes are the majority carriers.

OR

(b) Draw the circuit arrangement for studying the V-I characteristics of a p-n junction diode in forward bias and reverse bias. Show the plot of V-I characteristic of a silicon diode.

2

Ans: Circuit Arrangement

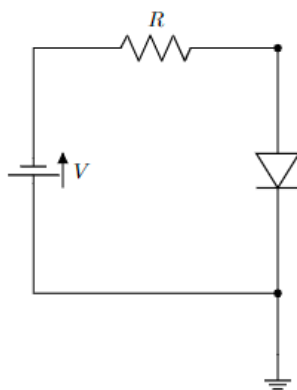
Forward Bias:

- Connect the positive terminal of the battery to the p-side (anode) of the diode.
- Connect the negative terminal of the battery to the n-side (cathode) of the diode.
- Include a resistor in series to limit the current.
- Measure the voltage across the diode and the current through the circuit.

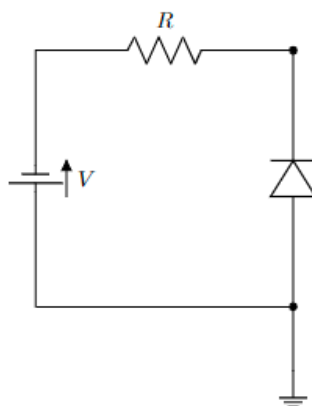
Reverse Bias:

- Connect the positive terminal of the battery to the n-side (cathode) of the diode.
- Connect the negative terminal of the battery to the p-side (anode) of the diode.
- Include a resistor in series to limit the current.
- Measure the voltage across the diode and the current through the circuit.

Circuit Arrangement for Forward Bias



Circuit Arrangement for Reverse Bias



25. Briefly explain how the diffusion and drift currents contribute to the formation of potential barrier in a p-n junction diode. 2

Ans:

Diffusion Current:

- **Process:** Electrons from the n-region diffuse into the p-region, and holes from the p-region diffuse into the n-region.
- **Effect:** This movement creates a depletion region with immobile ionized atoms, leading to a decrease in mobile charge carriers near the junction.

Drift Current:

- **Process:** The electric field created by the separation of charges in the depletion region causes remaining free electrons and holes to drift back to their original regions.
- **Effect:** This electric field opposes further diffusion, establishing a potential barrier.

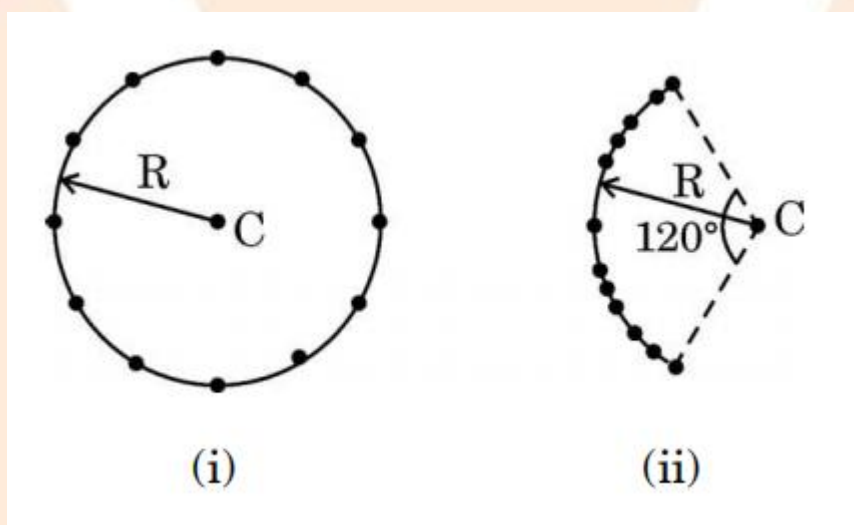
Potential Barrier:

- **Formation:** The balance between diffusion and drift currents creates a potential barrier, preventing free flow of charge carriers and establishing equilibrium.

SECTION C

26. (a) Twelve negative charges of same magnitude are equally spaced and fixed on the circumference of a circle of radius R as shown in Fig. (i).

Relative to potential being zero at infinity, find the electric potential and electric field at the centre C of the circle.



Ans: Given: Twelve negative charges of the same magnitude are equally spaced and fixed on the circumference of a circle of radius R .

Refer to Figure (i) for the arrangement of charges.

Electric Potential at the Centre C

The electric potential V due to a point charge q at a distance r is given by:

$$V = \frac{kq}{r}$$

where k is Coulomb's constant ($k = \frac{1}{4\pi\epsilon_0}$).

Since all charges are equally spaced and each charge is at a distance R from the centre:

$$V_i = \frac{kq}{R}$$

for each charge q .

There are 12 such charges, so the total potential at the centre C is the sum of the potentials due to each charge:

$$V_{\text{total}} = 12 \times \frac{kq}{R} = \frac{12kq}{R}$$

Electric Field at the Centre C

The electric field $\vec{E}$ due to a point charge q at a distance r is given by:

$$\vec{E} = \frac{kq}{r^2} \hat{r}$$

where $\hat{r}$ is the unit vector in the direction from the charge to the point of interest.

Since the charges are symmetrically placed, the electric field vectors due to each charge at the centre will cancel each other out. For every charge, there is an equal and opposite charge directly across the circle, resulting in their electric fields cancelling each other.

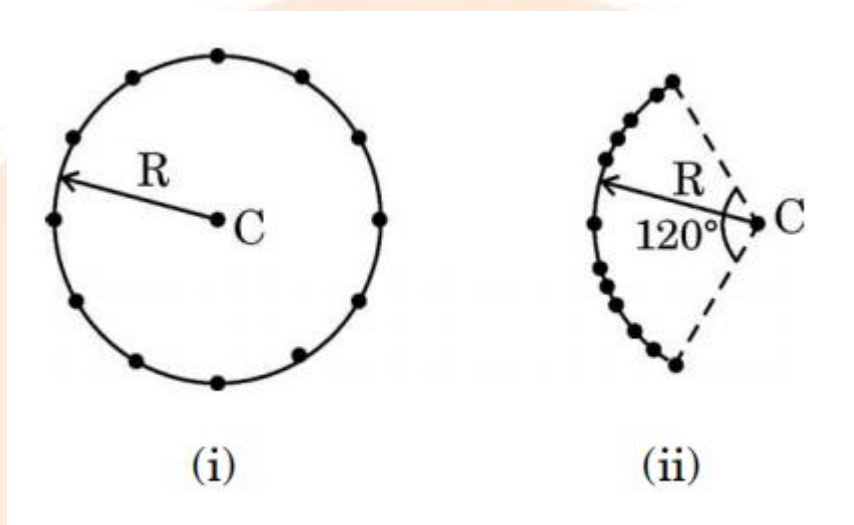
Thus, the net electric field at the centre C is:

$$\vec{E}_{\text{total}} = 0$$

- The electric potential at the centre C of the circle is: $V_{\text{total}} = \frac{12kq}{R}$

- The electric field at the centre (C) of the circle is: $\vec{E}_{\text{total}} = 0$

(b) If the charges are unequally spaced and fixed on an arc of 120° of radius R as shown in Fig. (ii), find electric potential at the centre C . 3



Ans: Given: Twelve negative charges of the same magnitude are unequally spaced and fixed on an arc of 120° of radius R as shown in Fig. (ii).

Electric Potential at the Centre C

The electric potential V due to a point charge q at a distance r is given by:

$$V = \frac{kq}{r}$$

where K is Coulomb's constant ($k = \frac{1}{4\pi\epsilon_0}$).

For a charge distribution along an arc, the distance of each charge from the centre C is R .

Since all charges are at the same distance R from the centre C , the potential due to each charge q at the centre is:

$$V_i = \frac{kq}{R}$$

If there are n charges placed along the arc, the total potential at the center C is the sum of the potentials due to each charge:

$$V_{\text{total}} = n \times \frac{kq}{R}$$

For the given problem, there are 12 charges along the arc of 120° . Therefore, the total potential at the centre C is:

$$V_{\text{total}} = 12 \times \frac{kq}{R} = \frac{12kq}{R}$$

- The electric potential at the centre C of the arc is:

$$V_{\text{total}} = \frac{12kq}{R}$$

27. (a) How does the resistance differ from impedance? With the help of a suitable phasor diagram, obtain an expression for impedance of a series LCR circuit, connected to a source $v = v_m \sin \omega t$. 3

Ans: Difference Between Resistance and Impedance:

- Resistance (R) is the opposition to DC current and is purely real. It does not change with frequency.
- Impedance (Z) is the opposition to AC current and includes both resistance and reactance, making it a complex quantity.

Impedance of a Series LCR Circuit:

In a series LCR circuit with source $v = v_m \sin(\omega t)$

- R is the resistance,

- L is the inductance,
- C is the capacitance,
- ω is the angular frequency,

The impedance Z is given by:

1. Phasor Diagram Analysis:

- Voltage across the resistor V_R is in phase with the current.
- Voltage across the inductor V_L leads the current by 90° .
- Voltage across the capacitor V_C lags the current by 90° .

2. Expression for Impedance:

$$Z = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

Where ωL is the inductive reactance, $\frac{1}{\omega C}$ is the capacitive reactance.

The magnitude of impedance is:

$$|Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

This shows the combined effect of resistance and reactance in the circuit.

OR

(b) Find the condition for resonance in a series LCR circuit connected to a source $v = v_m \sin \omega t$, where ω can be varied. Give the factors on which the resonant frequency of a series LCR circuit depends. Plot a graph showing the variation of electric current with frequency in a series LCR circuit. 3

Ans: Resonance in a Series LCR Circuit

Condition for Resonance:

Resonance occurs when the inductive reactance (X_L) equals the capacitive reactance (X_C).
The condition is:

$$\omega L = \frac{1}{\omega C}$$

Solving for resonant angular frequency ω_0 :

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

The resonant frequency f_0 is:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

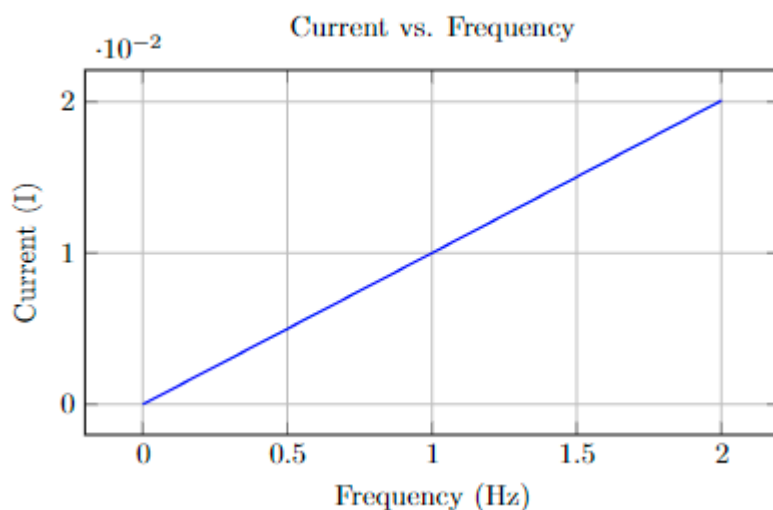
Factors Affecting Resonant Frequency:

1. Inductance (L): Increasing L lowers f_0
2. Capacitance (C): Increasing C also f_0 .

Graph of Electric Current vs. Frequency:

The graph shows:

- Current peaks at the resonant frequency f_0 .
- Current decreases on either side of f_0 due to increased impedance.



28. A long solenoid of radius r consists of n turns per unit length. A current $I = I_0 \sin \omega t$ flows in the solenoid. A coil of N turns is wound tightly around it near its centre. What is:

3

(a) the induced emf in the coil?

Ans:

1. Magnetic Field Inside the Solenoid:

$$B = \mu_0 n I = \mu_0 n I_0 \sin(\omega t)$$

2. Magnetic Flux Through the Coil:

The flux Φ through one turn of the coil is:

$$\Phi = B \cdot A = (\mu_0 n I_0 \sin(\omega t)) \cdot (\pi r^2)$$

3. Induced EMF:

Using Faraday's Law:

$$\text{emf} = -N \frac{d\Phi}{dt}$$

Compute the derivative:

$$\frac{d\Phi}{dt} = \mu_0 n I_0 \pi r^2 \omega \cos(\omega t)$$

Hence, the induced emf is:

$$\text{emf} = -N \mu_0 n I_0 \pi r^2 \omega \cos(\omega t)$$

(b) the mutual inductance between the solenoid and the coil?

Ans: The mutual inductance M is defined by the relation:

$$\Phi = MI$$

$$\text{Therefore: } M = \frac{\Phi}{I} = \frac{\mu_0 n I_0 \pi r^2 \sin(\omega t)}{I_0 \sin(\omega t)}$$

Simplifying:

$$M = \mu_0 n \pi r^2$$

29. How does Einstein's photoelectric equation explain the emission of electrons from a metal surface? Explain briefly. 3

Plot the variation of photocurrent with:

- (a) collector plate potential for different intensity of incident radiation, and**
- (b) intensity of incident radiation.**

Ans: Einstein's Photoelectric Equation and Its Explanation

Einstein's photoelectric equation describes how electrons are emitted from a metal surface when illuminated by light. The key points are:

- **Photoelectric Equation:** The kinetic energy (E_k) of the emitted electrons is given by:

$$E_k = h\nu - \phi$$

where h is Planck's constant, ν is the frequency of the incident light, and Φ is the work function of the metal.

- **Threshold Frequency:** Electrons are emitted only if the light's frequency is greater than a certain threshold frequency (ν_0). If $\nu < \nu_0$, no electrons are emitted regardless of the light's intensity.
- **Intensity:** The intensity of light affects the number of emitted electrons but not their energy.

Variation of Photocurrent

(a) Collector Plate Potential vs. Photocurrent for Different Intensities:

- **Collector Plate Potential:** The photocurrent increases with potential until it reaches a maximum, then decreases to zero as the potential becomes negative.

- **Different Intensities:** Higher intensity increases the maximum photocurrent, as more electrons are emitted.

(b) Intensity of Incident Radiation vs. Photocurrent:

- **Photocurrent vs. Intensity:** The photocurrent is directly proportional to the intensity of the incident light. Higher intensity results in a higher photocurrent.

30. (a) Draw the energy level diagram for hydrogen atom. Mark the transitions corresponding to the series lying in the ultraviolet region, visible region and infrared region. 3

Ans: Energy Level Diagram for Hydrogen Atom: The energy level diagram for a hydrogen atom illustrates the discrete energy levels of the electron around the nucleus. These levels are denoted by $n = 1, 2, 3, \dots$, where n is the principal quantum number. The energy level are given by:

$$E_n = -\frac{13.6\text{eV}}{n^2}$$

Energy Level Diagram for Hydrogen Atom

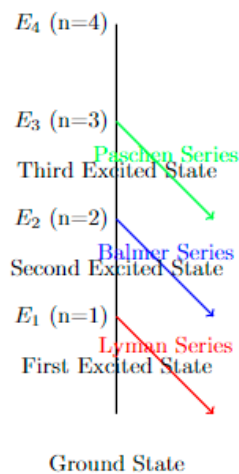


Figure 1: Energy Level Diagram for Hydrogen Atom. Transitions are marked for different spectral regions: Lyman Series (UV), Balmer Series (Visible), and Paschen Series (IR).

OR

(b) Draw a diagram to show the variation of binding energy per nucleon with mass number for different nuclei and mention its two features. Why do lighter nuclei usually undergo nuclear fusion? 3

Ans: Lighter nuclei, such as those of hydrogen and helium, usually undergo nuclear fusion because:

1. High Binding Energy per Nucleon: Lighter nuclei have lower binding energy per nucleon compared to nuclei around $A \approx 56$. By fusing lighter nuclei, the resulting heavier nucleus has a higher binding energy per nucleon, releasing energy in the process.

2. Energy Release: The fusion of lighter nuclei into heavier ones releases a significant amount of energy due to the conversion of mass into energy, as described by Einstein's equation $E = mc^2$. This energy release drives stellar processes and provides energy for stars and fusion reactors.

Variation of Binding Energy per Nucleon with Mass Number

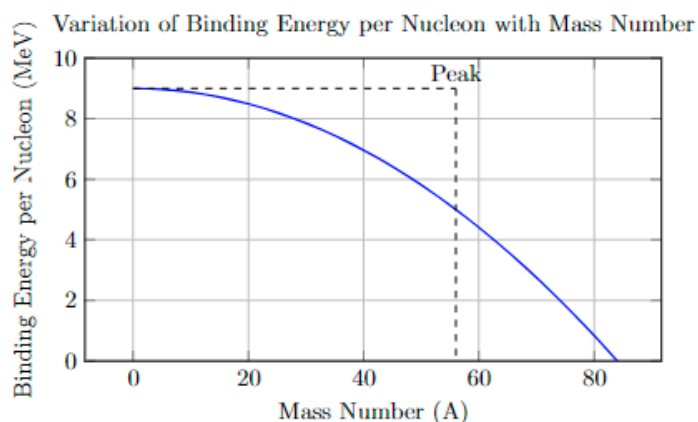


Figure 1: Variation of Binding Energy per Nucleon with Mass Number. The curve shows a peak around $A \approx 56$ and a general trend of decreasing binding energy for very heavy nuclei.

SECTION D

31. (a) (i) State Coulomb's law in electrostatics and write it in vector form, for two charges.

Ans: Statement: Coulomb's law states that the force between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them. The force acts along the line joining the charges.

Vector Form: For two point charges q_1 and q_2 , located at positions r_1 and r_2 respectively, the electrostatic force F on q_2 due to q_1 is given by:

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\mathbf{r}_2 - \mathbf{r}_1|^2} \hat{\mathbf{r}}$$

(ii) 'Gauss's law is based on the inverse-square dependence on distance contained in the Coulomb's law.' Explain.

Ans: Explanation:

Gauss's law states that the net electric flux through a closed surface is proportional to the enclosed electric charge. Mathematically, it is expressed as:

$$\oint_{\text{surface}} \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

where E is the electric field, dA is the differential area vector, and Q_{enc} is the total charge enclosed by the surface.

Connection to Coulomb's Law:

Gauss's law derives from Coulomb's law when considering the symmetry of charge distributions. Coulomb's law shows that the force between point charges follows an inverse-square law, which is reflected in the fact that the electric field E due to a point charge decreases with the square of the distance. Gauss's law incorporates this inverse-square dependency by integrating the electric field over a closed surface, yielding a proportionality to the enclosed charge, consistent with the behaviour described by Coulomb's law.

(iii) Two charges A (charge q) and B (charge $2q$) are located at points $(0,0)$ and (a, a) respectively. Let $\hat{i}$ and $\hat{j}$ be the unit vectors along x-axis and y-axis respectively. Find the force exerted by A on B, in terms of $\hat{i}$ and $\hat{j}$.

5

Ans:

1. Calculate the Vector Distance: $\mathbf{r}_{AB} = (a-0)\hat{i} + (a-0)\hat{j} = a\hat{i} + a\hat{j}$

2. Magnitude of the Distance: $|\mathbf{r}_{AB}| = \sqrt{a^2 + a^2} = \sqrt{2}a$

3. Unit Vector in the Direction of $\mathbf{r}_{AB}$: $\hat{\mathbf{r}}_{AB} = \frac{\mathbf{r}_{AB}}{|\mathbf{r}_{AB}|} = \frac{a\hat{i} + a\hat{j}}{\sqrt{2}a} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$

4. Force Calculation: $F = \frac{1}{4\pi\epsilon_0} \frac{q \cdot 2q}{|\mathbf{r}_{AB}|^2} = \frac{1}{4\pi\epsilon_0} \frac{2q^2}{2a^2} = \frac{q^2}{4\pi\epsilon_0 a^2}$

$$\mathbf{F} = \frac{q^2}{4\pi\epsilon_0 a^2} \cdot \frac{\hat{i} + \hat{j}}{\sqrt{2}} = \frac{q^2}{4\pi\epsilon_0 a^2 \sqrt{2}} (\hat{i} + \hat{j})$$

OR

(b) (i) Derive an expression for the electric field at a point on the equatorial plane of an electric dipole consisting of charges q and $-q$ separated by a distance $2a$.

Ans: 1. Electric Field Calculation:

The electric field E at a point on the equatorial plane of the dipole is given by: $E = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3}$

where p is the dipole moment: $p = q \cdot 2a$

Therefore: $E = \frac{1}{4\pi\epsilon_0} \frac{2qa}{r^3}$

This field is directed perpendicular to the dipole axis on the equatorial plane.

(ii) The distance of a far-off point on the equatorial plane of an electric dipole is halved. How will the electric field be affected for the dipole?

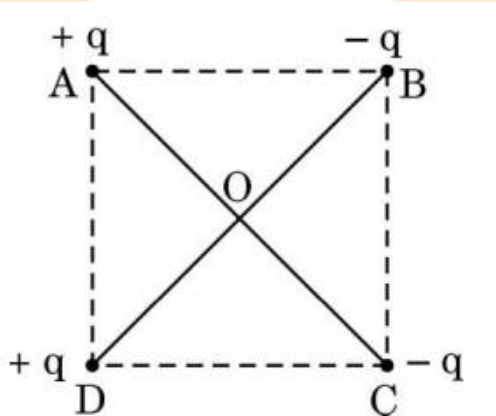
Ans: Electric Field Calculation at New Distance:

The new electric field E' is:
$$E' = \frac{1}{4\pi\epsilon_0} \frac{2qa}{\left(\frac{r}{2}\right)^3}$$

Simplify:
$$E' = \frac{1}{4\pi\epsilon_0} \frac{2qa}{\frac{r^3}{8}} = \frac{8}{4\pi\epsilon_0} \frac{qa}{r^3} = 8E$$

The electric field at the point will be increased by a factor of 8 when the distance is halved.

(iii) Two identical electric dipoles are placed along the diagonals of a square ABCD of side $\sqrt{2}$ m as shown in the figure. Obtain the magnitude and direction of the net electric field at the centre (O) of the square.



Ans:

1. Configuration:

- The dipoles are placed along the diagonals of the square.
- Let the dipole moment of each dipole be p .

2. Electric Field Due to One Dipole:

- At the centre O of the square, the distance from each dipole to $\frac{\sqrt{2}}{2} \cdot \sqrt{2} = 1\text{m}$
- The electric field E due to one dipole at O is given by:

$$E = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3}$$

Here, $r=1\text{ m}$, so:

$$E = \frac{1}{4\pi\epsilon_0} \frac{p}{1^3} = \frac{p}{4\pi\epsilon_0}$$

3. Direction of the Electric Field:

- For each dipole, the electric field will be directed along the axis of the dipole, away from the positive charge and towards the negative charge.
- Since the dipoles are oriented along the diagonals, their fields will add up vectorially.
- Due to symmetry, the components of the electric fields in opposite directions cancel out, leaving a net electric field along the diagonals.

4. Magnitude and Direction:

- The fields due to the dipoles cancel out in the direction perpendicular to the diagonals and add up along the diagonals.
- The resultant electric field at O is: $E_{net} = \frac{P}{4\pi\epsilon_0}$

The direction of E_{net} is along the diagonals of the square.

32. (a) (i) State Biot-Savart's law for the magnetic field due to a current carrying element. Use this law to obtain an expression for the magnetic field at the centre of a circular loop of radius 'a' and carrying a current 'I'. Draw the magnetic field lines for a current loop indicating the direction of magnetic field.

Ans:

(i) Biot-Savart's Law and Magnetic Field at the Centre of a Circular Loop

Biot-Savart's Law: Biot-Savart's Law states that the magnetic field dB at a point due to a small current element Idl is:

$$dB = \frac{\mu_0}{4\pi} \frac{Idl \times r}{r^3}$$

Magnetic Field at the Centre of a Circular Loop:

For a circular loop of radius a carrying a current I:

1. Each current element Idl contributes to the magnetic field at the centre. Due to symmetry, the contributions add up in the centre.

2. The total magnetic field B at the centre is: $B = \frac{\mu_0 I}{2a}$

Magnetic Field Lines:

- The magnetic field lines are concentric circles around the loop's plane.
- The direction follows the right-hand rule: fingers in the direction of current, thumb points in the field direction inside the loop.

(ii) Current Constituted by a Revolving Electron

- Speed of electron $v = 10^7$ m/s
- Radius of orbit $r = 10^{-10}$ m

1. Current Calculation:

Current I is the charge of the electron divided by the period T of the orbit:

$$I = \frac{e}{T}$$

2. Time Period T:

$$T = \frac{2\pi r}{v} = \frac{2\pi \times 10^{-10}}{10^7} = 2\pi \times 10^{-17} \text{ s}$$

3. Current:

$$I = \frac{1.6 \times 10^{-19}}{2\pi \times 10^{-17}} \approx 2.54 \times 10^{-3} \text{ A}$$

Thus, the current due to the revolving electron is approximately 2.54×10^{-3} A.

(ii) An electron is revolving around the nucleus in a circular orbit with a speed of 10^7 ms^{-1} . If the radius of the orbit is 10^{-10} , find the current constituted by the revolving electron in the orbit.

5

Ans: Current Calculation:

The electron moving in a circular orbit acts like a current loop. The current I can be calculated using the charge of the electron and the time period T of its orbit.

$$I = \frac{e}{T}$$

where e is the charge of the electron ($1.6 \times 10^{-9} \text{ C}$)

Calculate the Time Period T :

The time period T is the time taken for one complete revolution, which can be calculated from the circumference of the orbit and the speed of the electron.

$$T = \frac{2\pi r}{v}$$

Substitute $r = 10^{-10} \text{ m}$ and $v = 10^7 \text{ m/s}$:

$$T = \frac{2\pi \times 10^{-10}}{10^7}$$

$$T = 2\pi \times 10^{-17} \text{ s}$$

Calculate the Current I :

Substitute T into the formula for current:

$$I = \frac{e}{T}$$

$$I = \frac{1.6 \times 10^{-19}}{2\pi \times 10^{-17}}$$

$$I = \frac{1.6}{2\pi} \times 10^{-2}$$

$$I \approx 0.254 \times 10^{-2} \text{ A}$$

$$I \approx 2.54 \times 10^{-3} \text{ A}$$

Therefore, the current constituted by the revolving electron in the orbit is approximately $2.54 \times 10^{-3} \text{ A}$

OR

(b) (i) Derive an expression for the force acting on a current carrying straight conductor kept in a magnetic field. State the rule which is used to find the direction of this force. Give the condition under which this force is (1) maximum, and (2) minimum.

Ans: Expression for the Force:

When a straight conductor of length L carrying a current I is placed in a uniform magnetic field B , the force F on the conductor is given by the Lorentz force law:

$$F = I(L \times B)$$

where:

- F is the magnetic force,
- I is the current,
- L is the length vector of the conductor in the direction of the current,
- B is the magnetic field vector.

The magnitude of the force can be calculated as:

$$F = ILB\sin\theta$$

where θ is the angle between the direction of the current and the magnetic field.

Rule to Find the Direction of the Force:

The direction of the force is determined using the Fleming's Left-Hand Rule. According to this rule:

- The thumb of the left-hand points in the direction of the force,
- The index finger points in the direction of the magnetic field,
- The middle finger points in the direction of the current.

Conditions for Maximum and Minimum Force:

1. Maximum Force:

The force is maximum when the angle θ between the current and the magnetic field is

90° . In this case, $\sin\theta = 1$, so:

$$F_{\max} = ILB$$

2. Minimum Force:

The force is minimum (actually zero) when the angle θ is 0° or 180° . In these cases, $\sin\theta = 0$, so:

$$F_{\min} = 0$$

This occurs when the current direction is parallel or antiparallel to the magnetic field.

(ii) Two long parallel straight wires A and B are 2.5 cm apart in air. They carry 5.0A and 2.5A currents respectively in opposite directions. Calculate the magnitude of the force exerted by wire A on a 10 cm length of wire B. 5

Ans: The force per unit length f between two long parallel wires is given by:

$$f = \frac{\mu_0 I_A I_B}{2\pi d}$$

where μ_0 is the permeability of free space ($4\pi \times 10^{-7} \text{ H / m}$).

1. Calculate the force per unit length:

$$f = \frac{4\pi \times 10^{-7} \times 5.0 \times 2.5}{2\pi \times 0.025}$$

Simplify:

$$f = \frac{10 \times 10^{-7}}{0.025}$$

$$f = 4 \times 10^{-5} \text{ N/m}$$

2. Calculate the total force F for the given length L :

$$F = f \times L$$

$$F = 4 \times 10^{-5} \times 0.10$$

$$F = 4 \times 10^{-6} \text{ N}$$

The magnitude of the force exerted by wire A on a 10 cm length of wire B is $4 \times 10^{-6} \text{ N}$.

33. (a) (i) (1) Write two points of difference between an interference pattern and a diffraction pattern.

Ans: Nature of Pattern Formation:

- **Interference Pattern:** Formed by the superposition of coherent waves from multiple sources or slits, creating a series of equally spaced bright and dark fringes.
- **Diffraction Pattern:** Results from the bending of light waves around an obstacle or through a single slit, featuring a central bright maximum with decreasing intensity on either side, and less regular spacing compared to interference fringes.

Dependence on Experimental Setup:

- **Interference Pattern:** Fringe separation is influenced by the wavelength of light (λ) and the distance between the slits (d). For Young's double-slit experiment, the fringe width is given by $\Delta y = \frac{\lambda L}{d}$.
- **Diffraction Pattern:** The angular width of the central maximum and the spacing of diffraction minima depend on the slit width (a). For a single slit, the minima are found using $a \sin \theta = m\lambda$.

(2) Name any two factors on which the fringe width in a Young's double-slit experiment depends.

Ans:

- **Wavelength of Light (λ):** The fringe width increases with an increase in the wavelength of light.

- **Distance Between the Slits and the Screen (L):** The fringe width increases with an increase in the distance from the slits to the screen.

(ii) In Young's double-slit experiment, the two slits are separated by a distance equal to 100 times the wavelength of light that passes through the slits. Calculate :

(1) the angular separation in radians between the central maximum and the adjacent maximum.

Ans: The angular separation θ is given by:

$$\theta = \frac{\lambda}{d}$$

Substituting $d = 100\lambda$:

$$\theta = \frac{\lambda}{100\lambda} = \frac{1}{100} \text{ radians}$$

(2) the distance between these two maxima on a screen 50 cm from the slits. 5

Ans:

$$y = \frac{1}{100} \times 0.50 = 0.005 \text{ m} = 5 \text{ mm}$$

The linear distance y between the two maxima is:

$$y = \theta \times L$$

Substituting $\theta = \frac{1}{100}$ and $L = 0.50 \text{ m}$:

OR

(b) (i) A spherical surface of radius of curvature R separates two media of refractive indices n_1 and n_2 . A point object is placed in front of the surface at distance u in medium of refractive index n_1 and its image is formed by the surface at distance v , in the medium of refractive index n_2 . Derive a relation between u and v .

Ans: To derive the relation between u and v :

1. Refraction at the spherical surface:

The refraction formula for a spherical surface is:

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

2. Rearrange to find the relation between u and v :

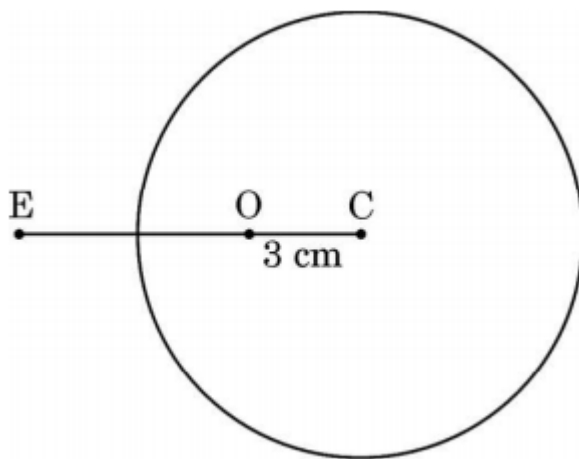
$$\frac{n_2}{v} = \frac{n_1}{u} + \frac{n_2 - n_1}{R}$$

Thus, the relation between u and v is:

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

(ii) A solid glass sphere of radius 6.0 cm has a small air bubble trapped at a distance 3.0 cm from its centre C as shown in the figure. The refractive index of the material of the sphere is 1.5. Find the apparent position of this bubble when seen through the surface of the sphere from an outside point E in air.

5



Ans: Formula:

The refraction formula at a spherical surface is:

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

Calculation:

$$\frac{1.0}{v} - \frac{1.5}{-3.0} = \frac{1.0 - 1.5}{6.0}$$

$$\frac{1.0}{v} + \frac{1.5}{3.0} = \frac{-0.5}{6.0}$$

$$\frac{1.0}{v} + 0.5 = -\frac{0.5}{6.0}$$

$$\frac{1.0}{v} = -\frac{1}{12} - 0.5$$

$$0.5 = \frac{1}{2}$$

$$\frac{1.0}{v} = -\frac{1}{12} - \frac{6}{12}$$

$$\frac{1.0}{v} = -\frac{7}{12}$$

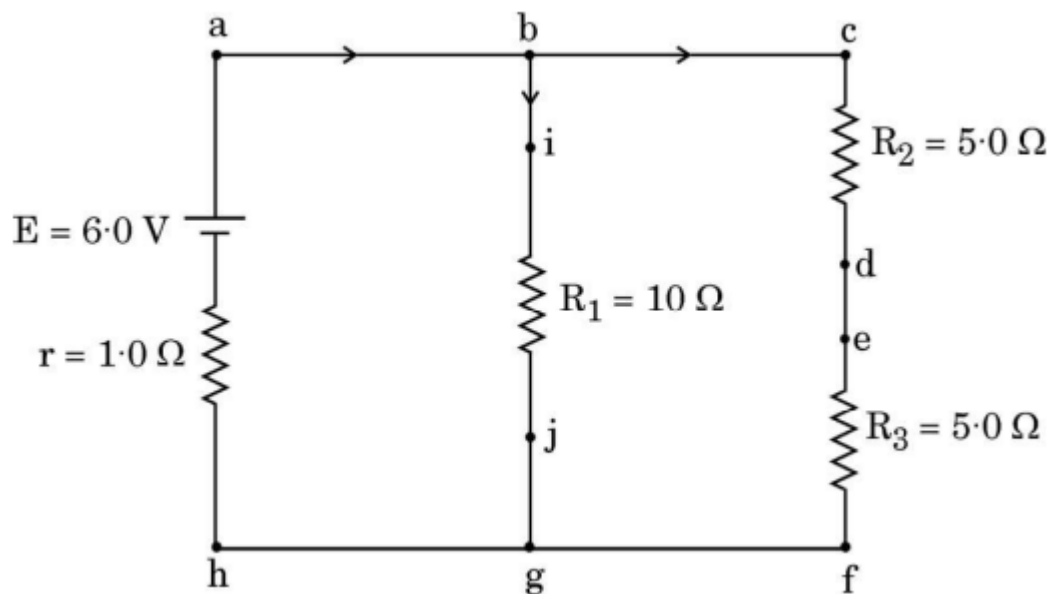
$$v = -\frac{12}{7} \text{ cm}$$

The apparent position of the bubble when seen through the surface of the sphere from an outside point E in air is: $v = -\frac{12}{7} \text{ cm} \approx -1.71 \text{ cm}$

The negative sign indicates that the image is formed on the same side of the surface as the object, meaning it appears closer to the observer. Thus, the bubble appears at approximately 1.71 cm inside the glass sphere from the surface.

SECTION E

34. The following figure shows a circuit diagram. We can find the currents through and potential differences across different resistors using Kirchhoff's rules.



Answer the following questions based on the above:

(a) Which points are at the same potential in the circuit?

1

Ans: b and i, e and f, g and j, h and a.

(b) What is the current through arm bg?

1

Ans: The current through arm bg is 0.444 A.

(c) Find the potential difference across resistance R_3 .

2

Ans: The potential difference across resistance R_3 is 1.11V.

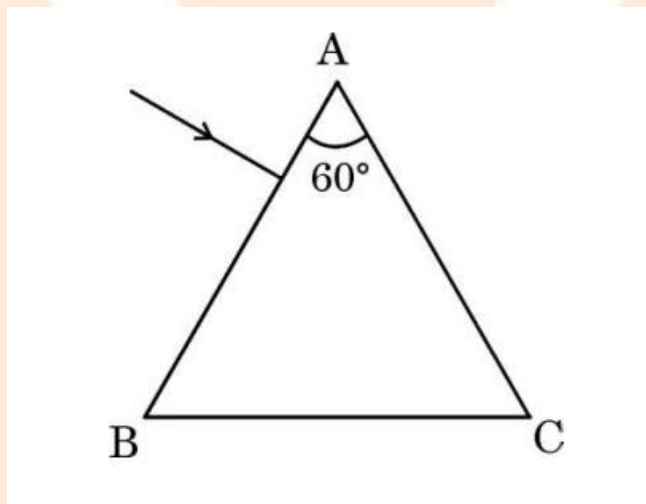
OR

(c) What is the power dissipated in resistance R_2 ?

2

Ans: The power dissipated in resistance R_2 is 0.2465 W.

35. Strontium titanate is a rare oxide - a natural mineral found in Siberia. It is used as a substitute for diamond because its refractive index and critical angle are 2.41 and 24.5° , respectively, which are approximately equal to the refractive index and critical angle of diamond. It has all the properties of diamond. Even an expert jeweller is unable to differentiate between diamond and strontium titanate. A ray of light is incident normally on one face of an equilateral triangular prism ABC made of strontium titanate.



Answer the following questions based on the above :

(a) Trace the path of the ray showing its passage through the prism.

1

Ans: When a ray of light is incident normally on face AB of the equilateral triangular prism ABC made of strontium titanate, it passes through the prism without deviation at AB. Upon reaching the face AC, the ray strikes at an angle of 60° , which is greater than the critical angle of 24.5° . Therefore, the ray undergoes total internal reflection at face AC and then again at face BC, eventually emerging out of the prism parallel to its original path.

(b) Find the velocity of light through the prism.

1

Ans: $v = \frac{3 \times 10^8 \text{ m/s}}{2.41} \approx 1.24 \times 10^8 \text{ m/s}$

(c) Briefly explain two applications of total internal reflection.

2

Ans:

1. Optical Fibers: Total internal reflection is the principle behind optical fibers. Light signals can travel long distances with minimal loss by reflecting internally within the fiber, making them essential for telecommunications and medical endoscopy.

2. Prism-based Instruments: Devices like periscopes and binoculars use prisms to reflect light internally, which allows for image direction changes without significant light loss or image degradation.

OR

(c) Define total internal reflection of light. Give two conditions for it.

2

Ans: Total internal reflection occurs when a light ray traveling in a denser medium strikes the boundary with a less dense medium at an angle greater than the critical angle, causing the light to be completely reflected back into the denser medium.

- Light must travel from a denser medium to a less dense medium.
- The angle of incidence in the denser medium must be greater than the critical angle for the two media.

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